

Predictive Vehicle Stability Assessment Using Lyapunov Exponent Under Extreme Conditions

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Abstract—Under extreme conditions, vehicles may encounter critical instability and cause traffic accidents due to the tire force saturation. In such cases, accurately predicting the vehicle instability is conducive to vehicle safety because drivers or vehicle controllers can be alerted and take early interventions to ensure driving safety. However, the existing stability assessment methods tend to be conservative, hard to quantify, and often ignore the coupled longitudinal and lateral dynamics, as well as the nonlinear characteristics of tires. Simultaneously, under extreme operating conditions, the assessment of vehicle potential risk imposes higher demands on the prediction accuracy of vehicle motion states. To address these 2 issues, this paper proposes a predictive vehicle stability assessment method using 3-dimensional Lyapunov exponents (3D-LEs) for a nonlinear vehicle system. Firstly, a nonlinear 8-degree-of-freedom vehicle dynamics model is constructed for an electric vehicle, aiming to capture the coupling dynamic characteristics and the tire force saturation under extreme conditions. To minimize the simulation-reality disparities, the vehicle parameters are automatically calibrated through Bayesian optimization using field test data. Secondly, to predict the potential risk of vehicle instability precisely, a physics-informed neural network based state prediction module is established for the vehicle stability assessment system. The ordinary differential equations of the vehicle system are integrated into neural networks to obtain physically consistent predictions of vehicle dynamic motion. Finally, the 3D-LEs, encompassing lateral motion, yaw motion, and roll motion, are employed to concurrently evaluate vehicle stability. Experimental results demonstrate that the predictive vehicle stability assessment method accurately evaluates the stability of predicted state sequences, enabling safer and more stable control under extreme conditions.

Index Terms—Extreme conditions, vehicle stability assessment, Lyapunov exponent, physics-informed neural network, vehicle state prediction.

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I. INTRODUCTION

WHEN driving on slippery, ice-snow roads, or during emergency steering or braking, as well as other extreme driving conditions, a vehicle may experience critical instability due to tire force saturation [1]. In these cases, the tires operate in highly nonlinear regions, impeding the vehicle from meeting the driver's intentions [2]. This often results in sideslips, drifting, rollovers, and other perilous situations. Consequently, enhancing vehicle safety has emerged as a focal point in the automotive industry's research endeavors.

At present, the technical challenges under extreme conditions mainly consist of modelling, cooperative motion control, and testing and evaluation. The testing and evaluation for vehicle motion control system under extreme conditions is essential for the automotive development and application. Generally, the real-world testing and evaluation under extreme conditions require multiple driving cycles and scenarios that are expensive, hard to reproduce and dangerous. Therefore, preliminary theoretical research and simulation can greatly reduce the cost and risk of real-vehicle testing and evaluation.

A comprehensive and accurate evaluation method of vehicle dynamics stability is critical for the effective deployment of vehicle safety control systems and is conducive to accelerating the vehicle testing under critical driving conditions [3]. The existing methods for assessing vehicle handling stability mainly encompass critical speed analysis, Lyapunov based method, phase plane method, and tire force estimation method [4]. The critical speed analysis method is based on the linear dynamic equation of vehicle system, and the stability is determined by the dynamic response of state variables, such as yaw rate and sideslip angle [5].

Lyapunov based methods mainly consist of the Lyapunov function based method and the Lyapunov exponent based method [6]. Lyapunov function based method is not limited to local stability, and does not explicitly solve dynamic equations. Instead, system stability is determined by constructing a scalar “energy like” function, namely the Lyapunov function. However, determining the Lyapunov function of complex systems is difficult, which hinders the practical application. Based on a 2-degree-of-freedom (DOF) nonlinear vehicle model, Johnson et al. constructed 2 Lyapunov functions using standard and modified kinetic energy functions, respectively. With these functions, they analyzed the vehicle lateral stability and determined a conservative elliptical stability region [7]. Unlike the Lyapunov function, the Lyapunov exponent represents the

numerical characteristics of the average exponential divergence rate of adjacent trajectories in phase space, that is, adjacent two points during the iteration process. Specifically, a negative exponent represents the contraction of the trajectory, corresponding to the stable state of the system. A positive exponent represents the separation of orbits, corresponding to the unstable state, and zero indicates the critical state. S. Sadri et al. first introduced the Lyapunov exponent to evaluate the vehicle stability. The results indicate that the rate at which the vehicle system returns to a stable region after being disturbed can be characterized by the maximum Lyapunov exponent, and the stability can be determined by the positive or negative Lyapunov exponent [8]. Subsequently, Shi et al. compared the differences between vehicle models with 2-DOF, 3-DOF, and 5-DOF, further verifying the feasibility of Lyapunov exponent based method [9].

The phase plane method generates the trajectories by the numerical integration of system equations in the time domain, and then estimate the stable region. It contains three types of phase planes: sideslip angle—sideslip angular velocity phase plane, sideslip angle—yaw rate phase plane, and slip angle of front tires—slip angle of rear tires phase plane. Inagaki et al. analyzed the influence of counter-steering and excessive steering on vehicle stability using the sideslip angle—sideslip angular velocity phase plane, and validated its effectiveness through the direct yaw control (DYC) algorithm [10]. Bobier Thiu et al. used the sideslip angle—yaw rate phase plane to determine the vehicle stability boundary. They applied it in developing an envelope control algorithm for enhancing yaw stability, as well as a sliding surface controller to stabilize the saddle point equilibrium in drifting [11]. Li et al. employed the front-tire slip angle—rear-tire slip angle phase plane as an evaluation index for a sliding mode controller to optimize the DYC in electric vehicles equipped with four in-wheel motors. This approach markedly enhanced vehicle handling and stability [12].

The precise acquisition of tire forces in real time is essential for evaluating vehicle stability. Tire force estimation methods can be classified into two approaches: observer-based and sensor-based methods [13]. Observer-based methods utilize tire models and vehicle motion states to estimate tire forces, relying on the tuning of model parameters under different tire loads [14]. For sensor-based methods, Xu et al. designed an intelligent tire system equipped with a tri-axial acceleration sensor and used neural networks to predict tire forces under different driving conditions [15]. Ding et al. proposed a comprehensive vehicle stability assessment system based on tire force estimation using low-cost on-board sensors [3].

Under extreme operating conditions, vehicles may encounter critical instability due to the tire force saturation. In such cases, accurately predicting the vehicle instability is conducive to vehicle safety because drivers or vehicle controllers can be alerted and take early interventions to ensure driving safety [16], [17]. Therefore, conducting multi-step prediction of vehicle dynamic motion is essential for assessing the potential vehicle instability under extreme conditions.

There are 3 main methods for predicting the dynamic motion of vehicle: physics-based methods, purely data-driven methods, and hybrid physics and neural network methods. For physics-based prediction methods, there are 2 major problems: 1) many properties of vehicle systems, especially nonlinear characteristics, cannot be explicitly modeled; and 2) the high-dimensional and nonlinear vehicle dynamics model generally consists of numerous parameters for describing physical characteristics [18]. The unmodeled dynamics and inaccuracy model parameters will lead to the rapid growth in the prediction error over time [19]. The rapid development of deep learning has addressed certain issues of physics-based prediction methods in diverse fields, including traffic flow prediction, vehicle safety prediction, and natural language processing, owing to its nonlinear approximation capability [20], [21], [22], [23]. However, the unavailability of big data and the physically inconsistent or implausible predictions hinder its application in forecasting the dynamics of vehicle system. In addition, the interpretability of neural networks is indispensable for some prediction systems that require high reliability [24]. Therefore, there is a pressing need to integrate physical laws or domain knowledge into neural network to enhance the predictive performance of learning algorithms. For vehicle dynamics system, there still exists a vast amount of prior knowledge that has not been used in neural networks. Embedding these physical principles into neural networks can constrain the solution space that enables the learning algorithms to converge quickly and generalize well [25], [26]. In turn, the integration of an inaccurate physics-based vehicle model into the neural network may lead to poor convergence and prediction performance.

To sum up, the existing researches have the following limitations: 1) the aforementioned methods for analyzing vehicle stability are typically based on a single-track model, and often ignore the coupled nonlinear characteristics of vehicle dynamics. Additionally, the discrepancy in vehicle dynamic characteristics between simulation and reality affects the assessment accuracy. 2) these assessment methods tend to be conservative, hard to quantify, and not applicable to extreme operating conditions. 3) Accurately predicting vehicle motion states under extreme conditions is highly challenging due to the rapid changes in vehicle states. How to embed the physical laws into neural networks to enhance the multi-step ahead prediction accuracy of vehicle dynamic motions has not been fully explored.

To tackle these issues, this paper proposes a predictive vehicle stability assessment method using 3-dimensional Lyapunov exponents (3D-LEs) for an 8-DOF nonlinear vehicle system. These 3D-LEs, encompassing the lateral velocity, yaw rate, and roll rate, offer a comprehensive evaluation of vehicle stability. Additionally, to accurately predict potential risks and intervene early under extreme conditions, we develop a physics-informed neural network (PINN) based vehicle state prediction module within the vehicle stability assessment system.

The main contributions of this paper are as follows:

1) A 3D-LEs based vehicle stability assessment method is proposed for an 8-DOF nonlinear vehicle dynamics

TABLE I
VEHICLE PARAMETERS

Symbol	Description	Value [Units]
m	Vehicle mass	1670[kg]
m_s	Sprung mass	1460.5[kg]
R_w	Wheel radius	0.308[m]
I_x	Roll moment of inertia	765 [kg · m ²]
I_z	Yaw moment of inertia	3043.8 [kg · m ²]
I_w	Wheel moment of inertia	0.92 [kg · m ²]
l_f	Distance from center of gravity to front axle	1.34[m]
l_r	Distance from center of gravity to rear axle	1.36[m]
t_f	Track width of front axle	1.475[m]
t_r	Track width of rear axle	1.480[m]
$K_{\phi f}$	Roll stiffness of front axle	47398[N · m/rad]
$K_{\phi r}$	Roll stiffness of rear axle	37211[N · m/rad]
$C_{\phi f}$	Roll damping of front axle	2348[N · m · s/rad]
$C_{\phi r}$	Roll damping of rear axle	2623[N · m · s/rad]
i_{sw}	Steering ratio	22
μ	Friction coefficient	0.3

system, which considers the lateral motion, yaw motion, and roll motion concurrently. These indices derived from vehicle dynamics system are utilized to determine the vehicle stability region and exactly quantify the degree of stability from 3 dimensions.

2) To precisely predict the potential risk of vehicle instability, a PINN based state prediction module is established for the vehicle stability assessment system. With this warning information, the vehicle controller can be alerted and take early interventions to avoid or reduce driving risks under extreme conditions. In this module, the ordinary differential equations of vehicle system is integrated into neural networks to obtain physically consistent predictions of vehicle dynamic motion. In addition, the parameters of vehicle dynamics model are auto-tuned by Bayesian optimization to narrow the gap between simulation and reality, which guarantees the predictive performance and convergence performance of PINN.

3) The stability control module with the predictive stability assessment method can intervene earlier, ensuring safer and more stable control under extreme conditions.

The rest of this paper is organized as follows. Section II presents the modelling of vehicle dynamics system. Section III describes the predictive vehicle stability assessment system, including the auto-tuning method of vehicle parameters, the PINN based vehicle state prediction module, and the 3D-LEs based vehicle stability assessment method. Section IV presents the experimental settings and results. Finally, the conclusion is made in Section V.

II. VEHICLE DYNAMICS MODEL

In this paper, a front-wheel drive electric vehicle is used to investigate vehicle stability under extreme conditions. The vehicle is propelled by 2 electric motors, providing a total wheel power of 160kW. The main vehicle parameters are presented in table I.

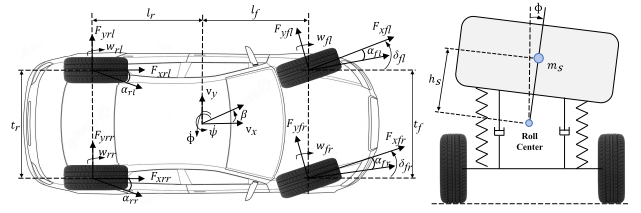


Fig. 1. 8-DOF vehicle dynamics model for a dual-motor electric vehicle.

A. Vehicle Model

Under extreme driving conditions, the highly coupled characteristics of vehicle dynamics cannot be ignored. Therefore, an 8-DOF vehicle dynamics model is constructed for the vehicle stability assessment, shown in Fig. 1. The degrees of freedom associated with the vehicle model include longitudinal motion, lateral motion, yaw motion, and roll motion, as well as four-wheel rotational motion. The dynamics equations governing the motion are derived as:

Longitudinal motion:

$$m(\dot{u} - \dot{\psi}v) + m_s h_s \ddot{\psi} \phi = (F_{xfl} + F_{xfr}) \cos \delta + F_{xrl} + F_{xrr} - (F_{yfl} + F_{yfr}) \sin \delta - F_{aero} - F_{roll} \quad (1)$$

Lateral motion:

$$m(\dot{v} + \dot{\psi}u) - m_s h_s \ddot{\phi} = (F_{yfl} + F_{yfr}) \cos \delta + F_{yrl} + F_{yrr} + (F_{xfl} + F_{xfr}) \sin \delta \quad (2)$$

Yaw motion:

$$I_z \ddot{\psi} = 0.5 t_r (F_{xrr} - F_{xrl}) - l_r (F_{yrl} + F_{yrr}) + 0.5 t_f \{ (F_{xfr} \cos \delta - F_{yfr} \sin \delta) - (F_{xfl} \cos \delta - F_{yfl} \sin \delta) \} + l_f \{ (F_{yfl} + F_{yfr}) \cos \delta + (F_{xfl} + F_{xfr}) \sin \delta \} \quad (3)$$

Roll motion:

$$I_x \ddot{\phi} - m_s h_s (\dot{v} + \dot{\psi}u) = m_s h_s g \phi - (C_{\phi f} + C_{\phi r}) \dot{\phi} - (K_{\phi f} + K_{\phi r}) \phi \quad (4)$$

Wheel rotational motion:

$$J_{wij} \dot{\omega}_{ij} = T_{dij} - T_{bij} - F_{xij} r_w - M_{ij} \quad i, j = fl, fr, rl, rr \quad (5)$$

where the subscripts, fl , fr , rl , rr , the front left, front right, rear left, rear right, respectively, m the vehicle mass, m_s the sprung mass, u the longitudinal velocity, v the lateral velocity, ψ the yaw angle, h_s the distance from body mass center to roll center, ϕ the roll angle, δ the steering angle, t_f and t_r the wheel track, l_f and l_r the distances from the center of mass to the front and rear axles respectively, I_x and I_z the moment of inertia in roll and yaw, F_x the longitudinal tire force, F_y the lateral tire force, F_{aero} the air resistance, F_{roll} the rolling resistance, C_{ϕ} and K_{ϕ} the equivalent damping coefficient and the lateral stiffness of suspension, the subscripts f and r the front and rear respectively, J_w the rotational inertia of tire, ω the angular velocity of tire, r_w the rolling radius, T_d the driving torque, T_b the braking torque, M the rolling resistance moment.

B. Tire Model

To analyze the nonlinear characteristics of vehicle system under extreme conditions, a semi-empirical model, namely Magic Formula (MF) tire model, is utilized. The general form of MF tire model can be expressed as:

$$\begin{cases} y = D \cdot \sin(C \cdot \arctan\{B \cdot x - E \cdot [x - \arctan(B \cdot x)]\}) \\ Y(X) = y(x) + S_V, \quad x = X + S_H \end{cases} \quad (6)$$

where Y is the output variable, including the longitudinal force F_x , the lateral force F_y or the aligning torque M_z . X is the slip angle α or the slip ratio λ , B is the stiffness factor, C is the shape factor (constant), D is the peak value, E is the curvature factor, S_H is the horizontal shift and S_V is the vertical shift. S_H and S_V are not considered in this paper. The calculation of B , D and E are:

$$\begin{cases} D = \mu(a_1 F_z^2 + a_2 F_z) \\ B_1 C_1 D_1 = (a_3 F_z^2 + a_4 F_z)(e^{-a_5 F_z}) \\ B_2 C_2 D_2 = a_3 \sin(a_4 \arctan(a_5 F_z)) \\ B = BCD/CD \\ E = a_6 F_z^2 + a_7 F_z + a_8 \end{cases} \quad (7)$$

where F_z is the vertical load of vehicle, $B_1 C_1 D_1$ is the stiffness of the lateral tire force, and $B_2 C_2 D_2$ is the stiffness of both longitudinal force (longitudinal slip stiffness) and self-aligning torque. The coefficient of tire formula a_i is obtained from [27].

The pure longitudinal F_{x0} and lateral F_{y0} tire forces in pure conditions can be obtained using equation 6. Considering the mutual coupling of longitudinal and lateral tire forces, the F_x and F_y in combined slip conditions can be expressed as [28]:

$$\begin{cases} \sigma_x = \lambda/(1 + \lambda) \\ \sigma_y = \tan \alpha/(1 + \lambda) \\ \sigma = \sqrt{\sigma_x^2 + \sigma_y^2} \\ F_x = (\sigma_x/\sigma) F_{x0}(\sigma) \\ F_y = (\sigma_y/\sigma) F_{y0}(\sigma) \end{cases} \quad (8)$$

where λ is the slip ratio, α is the slip angle, σ is the theoretical slip.

III. METHODOLOGY

A. Predictive Vehicle Stability Assessment System

In this paper, a predictive vehicle stability assessment system for an 8-DOF nonlinear vehicle system is proposed to predict the potential driving risk under extreme conditions. The vehicle stability assessment system is composed of 4 parts: a vehicle controller, an auto-tuned 8-DOF vehicle dynamics model for the electric vehicle, a physics-informed neural network (PINN) based vehicle state prediction method, and a 3-dimensional Lyapunov exponents (3D-LEs) based vehicle stability assessment method. This stability assessment system is illustrated in Fig. 2, and the procedure is shown in algorithm 1.

Algorithm 1 Predictive Vehicle Stability Assessment System Using 3-Dimensional Lyapunov Exponents

Input: historical sequences of vehicle states $S_{t-n:t} = [v_x, v_y, \dot{\psi}, \delta]$

- 1: Physics-informed based state prediction module: predicted sequences $S_{t+1:t+n} = [v_x, v_y, \dot{\psi}, \delta]$
- 2: **for** initial state $x_0 = s_{t+1}, s_{t+2}, \dots$ **do**
- 3: **Initialization:** initial vehicle states x_0 , initial state transient matrix: orthonormal vectors $[e_1^T(0) \cdots e_3^T(0)]_{3 \times 3}$, initial principal axes $\psi_0 = [e_1^T(0) \cdots e_3^T(0)]_{3 \times 3}$
- 4: **for** time step $t = 0, 1, \dots$ **do**
- 5: Calculate the 3×3 Jacobian matrix: $F_{ij}(t)$
- 6: Time evolution of principal axes: $\dot{\psi}_t = F(t)\psi_t$
- 7: Calculate the new orthonormal set through Gram-Schmidt reorthonormalisation method: $[e_1^T(t+1) \cdots e_3^T(t+1)]_{3 \times 3}$
- 8: Simulate the time evolution of a tuned vehicle dynamics system to generate state profiles: $\dot{x} = f(x)$
- 9: **end for**
- 10: Calculate the 3-dimensional Lyapunov exponents based on the simulated state profiles: $\lambda_i (i = 1, 2, 3)$
- 11: **end for**

Output: all the 3-dimensional Lyapunov exponents: lateral motion, yaw motion, roll motion

Specifically, the vehicle controller is designed to generate driving torque or braking torque using a proportional-integral-derivative (PID) algorithm to track the target speed. A stability control module, implemented by sliding mode control, is integrated into the vehicle system to enhance vehicle stability under extreme conditions. With the control commands, a nonlinear 8-DOF vehicle dynamics model is established for execution and simulation. Given the significance of precise vehicle dynamics modelling for state prediction and vehicle stability assessment, Bayesian optimization is utilized to automatically calibrate vehicle parameters, reducing the disparity between simulation and reality. The physical information obtained from this calibrated vehicle model is subsequently embedded into neural networks to achieve multi-step vehicle state prediction using PINN. In the 3D-LEs based vehicle stability assessment method, each set of vehicle states from the predicted sequences $S_{t+1:t+n}$ serves as the initial state to simulate the time evolution of the tuned vehicle dynamics system, thereby generating state profiles. Based on these profiles, the 3D-LEs, including lateral motion, yaw motion, and roll motion, are calculated to assess the potential risk of vehicle instability under extreme conditions.

B. Vehicle Parameter Auto-Tuning

Although more complex physics-based vehicle dynamics models can better approximate real vehicle behavior, their accuracy depends heavily on parameterization [29]. Typical parameters in these models include stiffness, inertia, coefficients of the mechanical system, as well as tire pressure, temperature, and wear condition. However, these parameters are not easily measured or identified for highly complex and

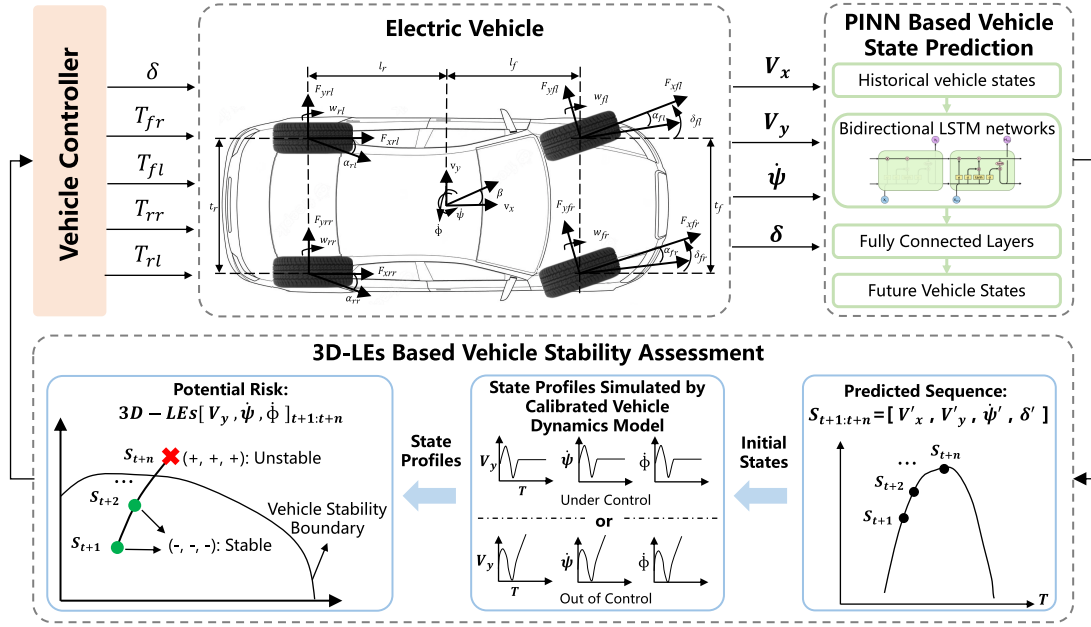


Fig. 2. Predictive vehicle stability assessment system using 3-dimensional Lyapunov exponents.

coupled vehicle systems. Therefore, a global optimization method, namely Bayesian optimization, is utilized to bridge the gap between simulated vehicle models and real vehicles. An accurate vehicle dynamics model can significantly enhance the prediction performance of vehicle motion states and improve the accuracy of vehicle stability assessment.

Bayesian optimization is a machine learning approach used for finding the extrema of functions that are computationally expensive [30]. It is particularly useful for functions where calculating derivatives is challenging and when the learning algorithms involve numerous hyperparameters [31]. The core principle of Bayesian optimization is to construct a surrogate model to approximate the unknown cost function. This surrogate model guides the selection of new evaluation points through the design of an acquisition function [32].

The tuning of vehicle dynamics model is implemented by adjusting environmental and vehicle parameters, including friction coefficient, rolling resistance coefficient, driving torque, and tire parameters, etc. Equation 7 specifies the coefficients a_i as the tuning parameters for the lateral and longitudinal tire forces. These parameters are determined based on inputs involving driving torque and steering angle. The objective function aims to optimize the adjustable vehicle or environmental parameters θ of the mathematical model, minimizing the modeling errors compared to the real vehicle. The objective function can be expressed as:

$$J = \frac{1}{N} \left\| \begin{bmatrix} \dot{\psi} \\ v_x \\ v_y \end{bmatrix}_i - \begin{bmatrix} \hat{\dot{\psi}} \\ \hat{v}_x \\ \hat{v}_y \end{bmatrix}_i \right\| \quad (9)$$

where v_x , v_y and $\dot{\psi}$ are the outputs of the mathematical model, $\hat{v}_x$, $\hat{v}_y$ and $\hat{\dot{\psi}}$ are the measured data. The modeling errors of longitudinal velocity, lateral velocity, and yaw rate serve as the evaluation indices to obtain the optimal parameters θ^* .

The details of this vehicle parameter auto-tuning method through Bayesian optimization can be seen in [33].

C. Physics-Informed Neural Network Based Vehicle State Prediction

Modeling and predicting the evolution of a nonlinear and high-dimensional system remain severe challenges due to the noisy data collected and the complexity of systems characterized by parameterized ordinary differential equations (ODEs). Although machine learning has demonstrated a promising method in addressing such challenges, it typically requires large amounts of data for training neural networks, which may not always be available. To solve these limitations, PINN integrates information from both measured data and ODEs by embedding the ODEs into the loss function of a neural network through automatic differentiation [34]. Unlike fully data-driven algorithms, PINN can enforce physical invariants to enhance prediction accuracy, improve generalization, and accelerate training speed.

To predict potential vehicle risks, a PINN based vehicle state prediction method is developed for the vehicle stability assessment system. This enables early intervention by the vehicle controller when encountering extreme driving conditions, thereby enhancing vehicle safety and handling stability. The PINN based vehicle state prediction consists of 2 parts: vehicle dynamics equations expressed by ODEs and a deep neural network. Vehicle dynamics characteristics from 3 ODEs, including longitudinal motion, lateral motion and yaw motion, are embedded into the deep neural network. The deep neural network, implemented by the bidirectional long short-term memory (LSTM), can process the input sequences from both positive and negative time directions simultaneously [35]. The framework of PINN based vehicle state prediction is illustrated in Fig. 3.

The inputs of PINN consist of 4 features: the historical 5 time steps of longitudinal velocity, lateral velocity, yaw rate and steering angle. The outputs of PINN are to predict the motion states for the next 5 time steps. The loss function $\mathcal{L}$ is composed of 2 parts: a supervised loss of data measurements $\mathcal{L}_{\text{data}}$ and an unsupervised loss of ODEs $\mathcal{L}_{\text{ODE}}$.

$$\mathcal{L} = w_{\text{data}}\mathcal{L}_{\text{data}} + w_{\text{ODE}}\mathcal{L}_{\text{ODE}} \quad (10)$$

where w_{data} and w_{ODE} are the weights used to balance between the supervised loss of measured data and the unsupervised loss of ODEs. The weights, w_{data} and w_{ODE} , are set as 0.8 and 0.2, respectively, based on the evaluated prediction performance of vehicle motion states under different weight settings.

$$\mathcal{L}_{\text{data}} = \frac{1}{N_{\text{data}}} \sum_{i=1}^{N_{\text{data}}} (y_i - \hat{y}_i)^2 \quad (11)$$

where y is the predictive value, $\hat{y}$ is the label from measured data,

$$\mathcal{L}_{\text{ODE}} = \frac{1}{N_{\text{ODE}}} \sum_{j=1}^{N_{\text{ODE}}} \left(f_j(t, x) - \frac{dy_j}{dt} \right)^2 \quad (12)$$

where $f_j(t, x)$ is the ODEs of vehicle dynamics system.

D. Lyapunov Exponent

For a continuous dynamical system in an n -dimensional phase space, the evolution of each dimension can be represented by an n -dimensional ellipsoid. The Lyapunov exponent of such a system is a quantity that characterizes the average exponential rates of divergence or convergence of nearby orbits in phase space [36]. The signs of Lyapunov exponents represent the stability properties of a nonlinear dynamic system. Negative values indicate that the principal axes of an n -ellipsoid shrink on average, whereas positive values correspond to chaotic behavior within the nonlinear system.

The vehicle dynamics system is a dissipative nonlinear system due to the resistance experienced by the vehicle and the highly nonlinear properties of tires. Therefore, the Lyapunov exponent is applicable in vehicle systems to identify chaotic behavior [8].

For an n -dimensional vehicle dynamics system, the spectrum of Lyapunov exponents is denoted as $\{\lambda_1, \lambda_2, \dots, \lambda_n\}$. The i th Lyapunov exponent λ_i can be calculated through the length of the principal axis $P_i(t)$ of an n -ellipsoid:

$$\lambda_i = \lim_{t \rightarrow \infty} \frac{1}{t} \log \frac{P_i(t)}{P_i(t_0)}, \quad i = 1, 2, \dots, n \quad (13)$$

where n corresponds to the dimension of dynamical system. Different Lyapunov exponents represent the expanding or contracting properties of their respective dimensions in phase space. In finite time, all these Lyapunov exponents will converge to their true values.

Based on the definition of Lyapunov exponent, a standard algorithm [36] is applied to calculate the Lyapunov exponent of a dynamical system, which is presented in the form of mathematical model as follows.

The time evolution of a dynamical system can be described by a set of differential equations in M -dimensional Euclidean state space R^m as:

$$\dot{x} = f(x), \quad (14)$$

where f is an $n \times 1$ nonlinear vector function and x is an $n \times 1$ state vector. The evolution of principal axes is defined as:

$$\dot{\psi}_t = F(t)\psi_t \quad (15)$$

where ψ_t the state transient matrix of the linearised system $\delta x(t) = \psi_t \delta x(0)$ and $F(t)$ the $n \times n$ Jacobian matrix that is calculated by:

$$F_{ij}(t) = \left. \frac{\partial f_i}{\partial x_j} \right|_{x_j=x_j(t)} \quad (16)$$

The initial principal axes are set as orthonormal vectors, i.e., $e_1(0) = (1, 0, \dots, 0)$, $e_2(0) = (0, 1, 0, \dots, 0)$, $\dots$, $e_n(0) = (0, \dots, 0, 1)$. The initial conditions are given by:

$$\begin{Bmatrix} x(0) \\ \psi_t(0) \end{Bmatrix} = \begin{Bmatrix} x_0 \\ \psi_0 \end{Bmatrix}, \quad (17)$$

where the initial state transient matrix is defined based on the initial principal axes:

$$\psi_0 = [e_1^T(0) \cdots e_n^T(0)]_{n \times n} \quad (18)$$

Based on the equations 14, 15 and 17, the next set of vectors are generated with a time-step τ . In order to avoid misalignment of all the vectors along the direction of maximal expansion, the Gram-Schmidt reorthonormalisation (GSR) method is used to provide the new orthonormal set $\{e_1(\tau) \cdots e_n(\tau)\}$, (19) as shown at the bottom of the next page.

With finite amounts of time-series data, the Lyapunov exponent can be estimated as:

$$\lambda_i \approx \frac{\sum_{m=1}^s \log P_i(m)}{s\tau} \quad (i = 1, \dots, n) \quad (20)$$

where $\{P_1(m), P_2(m), \dots, P_n(m)\}$ the denominators of $\{e_1(\tau) \cdots e_n(\tau)\}$, s the number of integration step.

IV. EXPERIMENT AND RESULT

In this section, we first introduce the experimental datasets collected from the real vehicle on an ice-snow road and experimental settings. Subsequently, we analyze the performance of physics-informed neural network (PINN) based vehicle state prediction method compared to a purely data-driven method. In addition, the vehicle stability boundaries for a front-wheel drive electric vehicle under different scenarios are determined and analyzed using 3-dimensional Lyapunov exponents (3D-LEs). Finally, we validate the predictive vehicle stability assessment method using a collection of typical testing scenarios under extreme conditions.

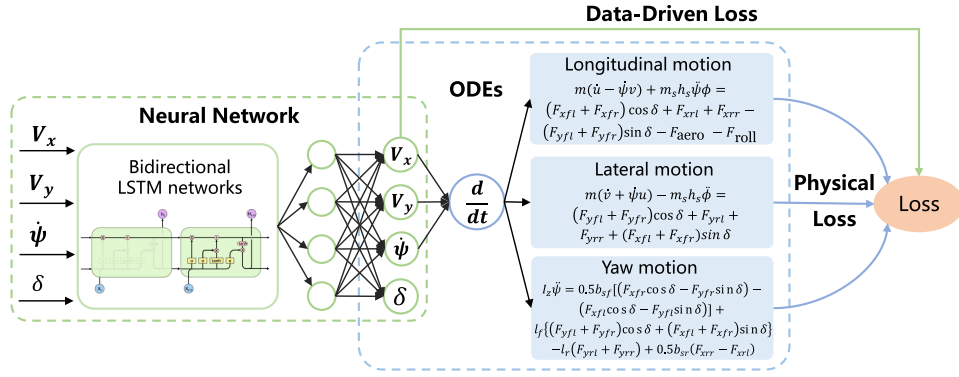


Fig. 3. Physics-informed neural network based vehicle state prediction.

A. Experimental Data and Settings

The field test data under extreme conditions are collected from an integrated inertial navigation system and a controller area network (CAN) bus on ice-snow roads, capturing velocity, acceleration, yaw rate, steering angle, driving torque, braking torque, etc. These data are recorded at a sampling rate of 100 Hz.

The testing scenarios involve the evasive maneuver (EM) test and the pylon course slalom (PCS) test, which are typical procedures for testing the handling and stability of automobiles. Unlike the standard test procedures, these test procedures are conducted on an ice-snow road with a friction coefficient of 0.3. The evasive maneuver test assesses a vehicle's ability to evade a suddenly appearing obstacle by navigating an S-shaped course delineated by traffic cones. Fig. 4 and 5 illustrate the longitudinal speed and steering angle in these handling and stability test procedures.

In the PINN based vehicle state prediction module, we divide these measured data into training and testing sets in a 7:3 ratio. The prediction horizon length is 0.5s with a time step of 0.1s. Therefore, a 5-step ahead prediction is sufficient for the control module to intervene in advance.

In this paper, due to the absence of roll motion data collected from the real vehicle, neither the PINN based vehicle state prediction nor the parameter auto-calibration through Bayesian optimization utilizes the roll motion data.

B. Accuracy of Auto-Tuned Vehicle Model

The vehicle and environmental parameters are auto-tuned by Bayesian optimization through the field test data gathered under extreme conditions. The simulation accuracy of the calibrated vehicle dynamics model is measured by root mean

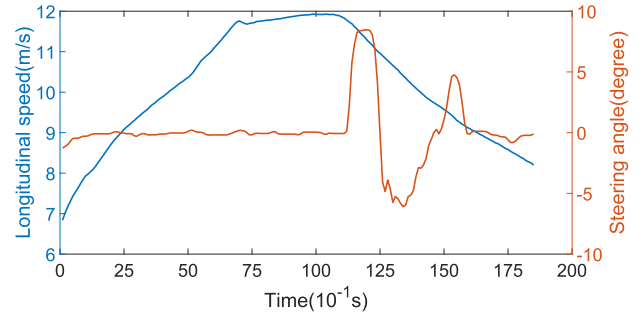


Fig. 4. Evasive maneuver test.

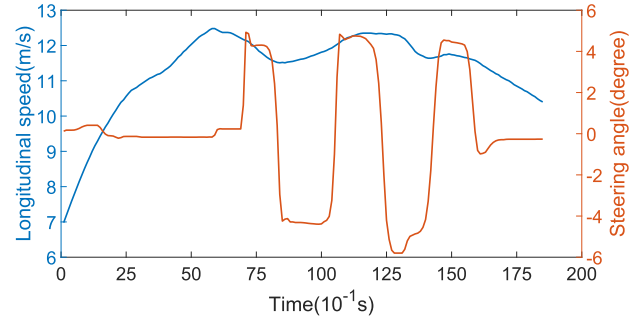


Fig. 5. Pylon course slalom test.

squared error (RMSE). For the EM tests in Fig. 6, 7 and 8, the average RMSEs of longitudinal speed, lateral speed, and yaw rate are 0.336 m/s, 0.066 m/s, and 0.022 rad/s, respectively. In the PCS tests, the average errors of these three motions in table II are 0.301 m/s, 0.197 m/s, and 0.032 rad/s, respectively.

$$\begin{cases} e_1(\tau) = \frac{v_1(\tau)}{\|v_1(\tau)\|} \\ e_2(\tau) = \frac{v_2(\tau) - \langle v_2(\tau), e_1(\tau) \rangle e_1(\tau)}{\|v_2(\tau) - \langle v_2(\tau), e_1(\tau) \rangle e_1(\tau)\|} \\ \dots \\ e_n(\tau) = \frac{v_n(\tau) - \langle v_n(\tau), e_1(\tau) \rangle e_1(\tau) - \dots - \langle v_n(\tau), e_{n-1}(\tau) \rangle e_{n-1}(\tau)}{\|v_n(\tau) - \langle v_n(\tau), e_1(\tau) \rangle e_1(\tau) - \dots - \langle v_n(\tau), e_{n-1}(\tau) \rangle e_{n-1}(\tau)\|} \end{cases} \quad (19)$$

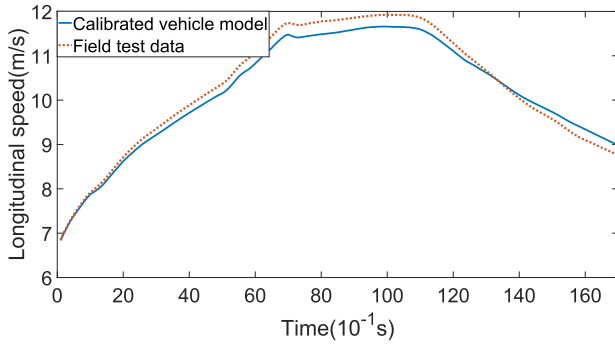


Fig. 6. Longitudinal speed of calibrated vehicle model and real vehicle.

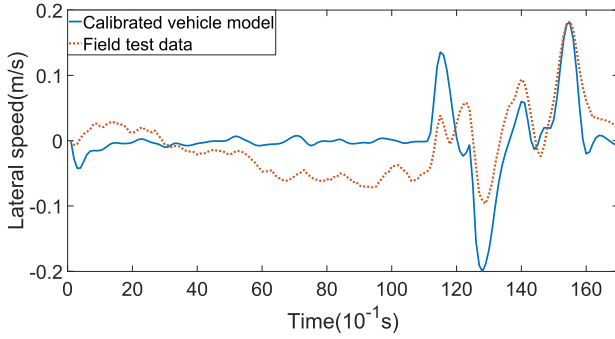


Fig. 7. Lateral speed of calibrated vehicle model and real vehicle.

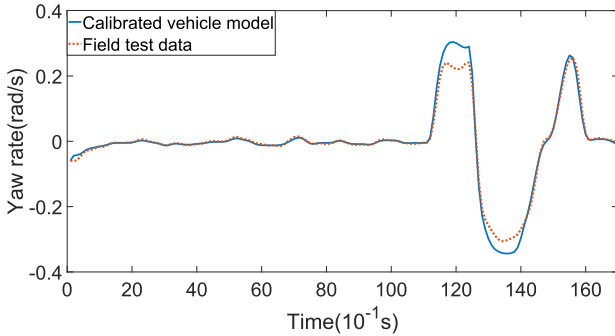


Fig. 8. Yaw rate of calibrated vehicle model and real vehicle.

TABLE II
ACCURACY OF CALIBRATED VEHICLE MODEL

Field test data	RMSE V_x (m/s)	RMSE V_y (m/s)	RMSE $\dot{\phi}$ (rad/s)
EM test 1	0.203	0.047	0.019
EM test 2	0.452	0.076	0.017
EM test 3	0.196	0.075	0.029
PCS test 1	0.353	0.187	0.033
PCS test 2	0.268	0.194	0.028
PCS test 3	0.283	0.210	0.036

C. PINN Based Vehicle State Prediction

With the historical information of vehicle states, PINN is utilized to generate the multi-step ahead forecasts of longitudinal motion, lateral motion, yaw motion, and steering angle. By incorporating the ordinary differential equations (ODEs) of

TABLE III
VEHICLE MOTION PREDICTION ERROR

Field test data	RMSE (PINN)	RMSE (Purely data-driven)
EM test 1	0.039	0.045
EM test 2	0.043	0.046
EM test 3	0.036	0.042
PCS test 1	0.041	0.048
PCS test 2	0.042	0.049
PCS test 3	0.039	0.046

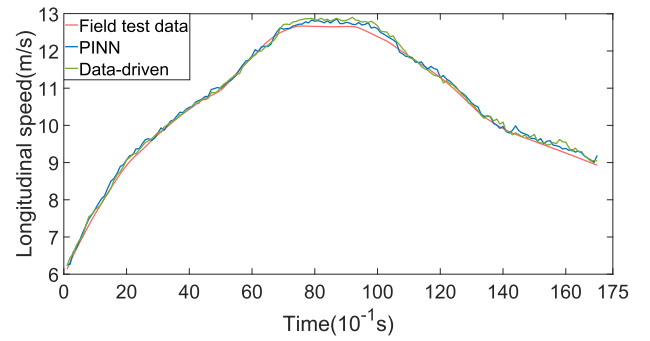


Fig. 9. Longitudinal motion prediction (EM test 3).

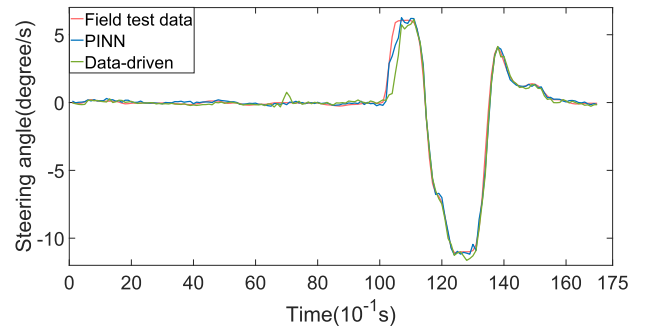


Fig. 10. Steering angle prediction (EM test 3).

vehicle dynamics model into the bidirectional long short-term memory network, the PINN based method outperforms the purely data-driven method in terms of prediction accuracy and physical consistency.

In contrast to the purely data-driven method, the prediction error (RMSE) of the PINN based method decreases by an average of 13.07%, shown in table III. Fig. 9, 10 and 11 illustrate the predicted profiles of longitudinal velocity, steering angle and yaw rate at the $t + 5$ time step. These two methods exhibit no significant differences in the prediction performance, which can be attributed to the relatively minor fluctuations observed in both motion profiles. Unlike the former two motion profiles, Fig. 12 demonstrates that the lateral velocity predicted by PINN is more physically consistent with the field test data compared to the purely data-driven method, resulting in a 20.78% reduction in the prediction error.

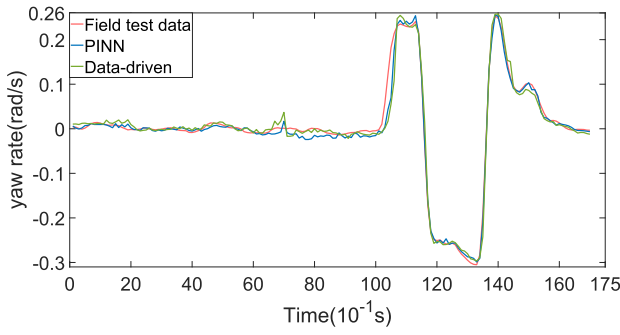


Fig. 11. Yaw motion prediction (EM test 3).

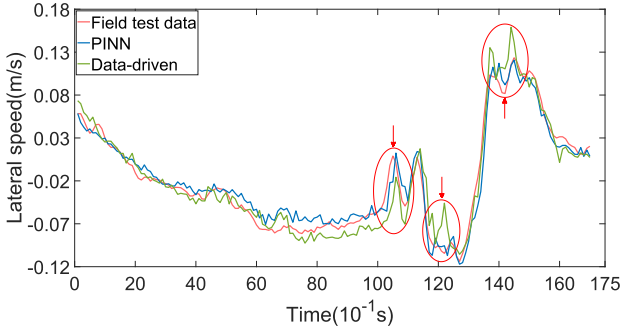


Fig. 12. Lateral motion prediction (EM test 3).

D. Vehicle Stability Boundary

Using the 3D-LEs based vehicle stability assessment method, the vehicle stability boundary can be determined by the signs of Lyapunov exponents. When all the 3D-LEs are negative, the vehicle system is stable. Conversely, a positive Lyapunov exponent in any dimension indicates instability in that corresponding dimension, which renders the entire vehicle system unstable. To demonstrate the accuracy of the 3D-LEs method, the phase plane method is introduced for comparison. The phase plane is generated using the same calibrated vehicle model. Experimental results indicate that the proposed method can not only precisely determine the vehicle stability boundary under extreme conditions but also quantitatively evaluate the degree of vehicle stability.

In the absence of steering angle input, the stability region demonstrates symmetry with respect to the origin. In Fig. 13, it can be observed that the vehicle stability region of the proposed method exhibits a high degree of overlap with the stability region determined by the phase plane method at a speed of 10m/s. In Fig. 14, a notable contraction of the stability region is observed with the escalation of speed, particularly evident in areas characterized by high yaw rates. Specifically, compared to the stability region at a longitudinal speed of 10m/s, a decrease of 14.33% is noted at 20m/s, followed by an additional reduction of 11.46% at a longitudinal speed of 30m/s, shown in Fig. 15.

When steering angle input is introduced, the lateral tire forces induced by steering lead to asymmetry in the stability region. Under the condition of a speed of 10m/s and a steering angle of 5° in Fig. 16, the stability region derived from the 3D-LEs method is highly consistent with that derived from

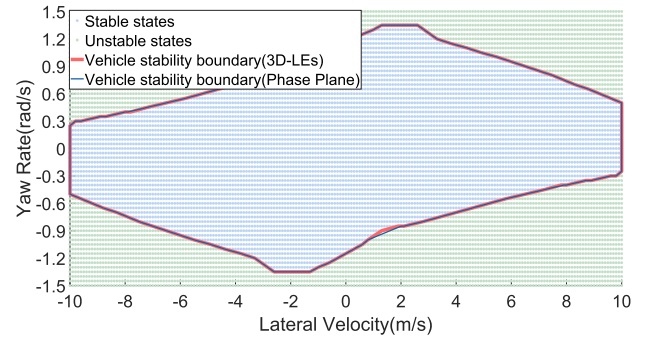


Fig. 13. Comparison of vehicle stability boundary of 3D-LEs and phase plane.

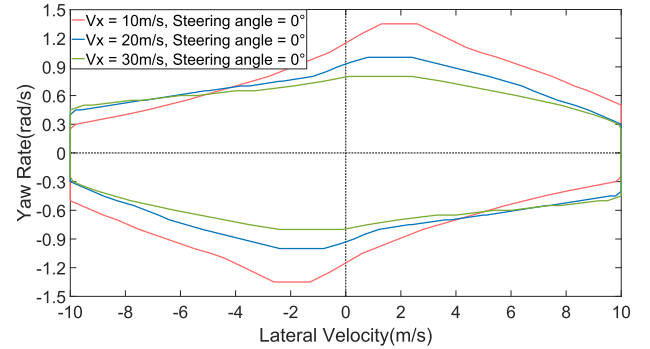


Fig. 14. Comparison of vehicle stability boundary under different velocities.

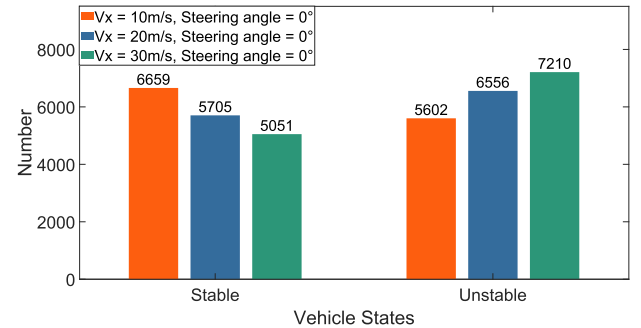


Fig. 15. Distributions of vehicle stability under different velocities.

the phase plane method. As the steering angle increases, the stability region shifts downwards as a whole, accompanied by the slight contraction of the stability region, as shown in Fig. 17 and 18. In addition, we further study the stability regions under variable longitudinal speeds and a fixed steering angle. Results in Fig. 19 illustrate that the stability region at a speed of 20m/s shrinks by 15.77% compared to that at a longitudinal speed of 10m/s. When further increasing to 30m/s, the stability region experiences a sharp decrease of 61.0%, particularly in the positive lateral velocity range.

E. Predictive Vehicle Stability Assessment Under Extreme Conditions

With the PINN based state prediction and the established vehicle stability boundary, we further validate the predictive vehicle stability assessment system using the field test data collected under extreme conditions, including EM and PCS

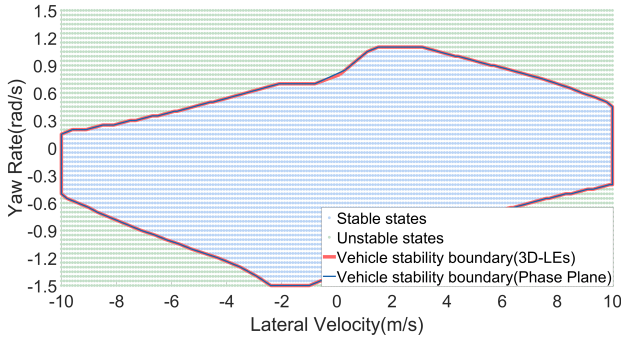


Fig. 16. Comparison of vehicle stability boundary of 3D-LEs and phase plane.

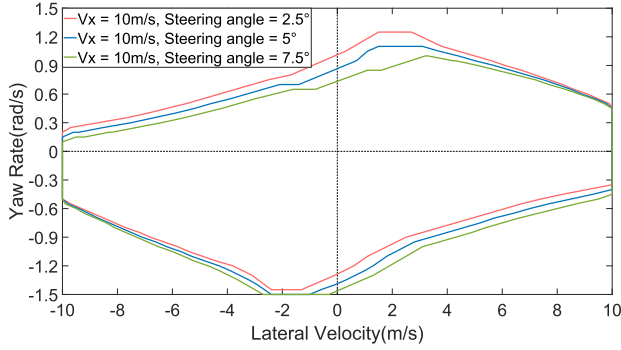


Fig. 17. Comparison of vehicle stability boundary under different steering angles.

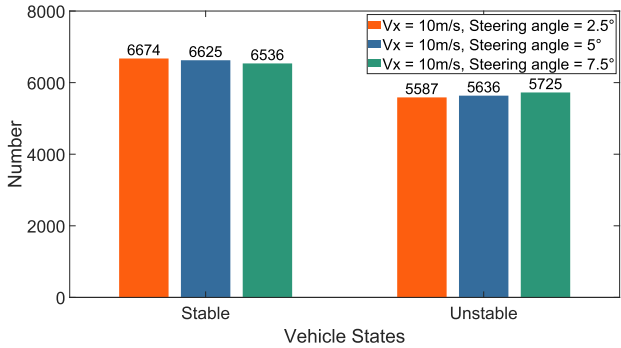


Fig. 18. Distributions of vehicle stability under different steering angles.

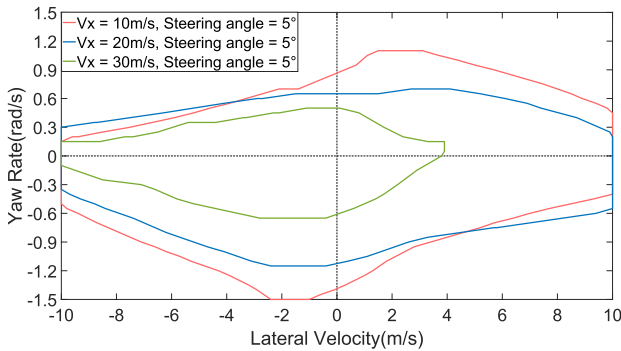


Fig. 19. Comparison of vehicle stability boundary under different velocities with steering angle input.

tests. In Fig. 21 and 22, we select 4 vehicle state prediction sequences (A-D), characterized by quick turns at high speeds

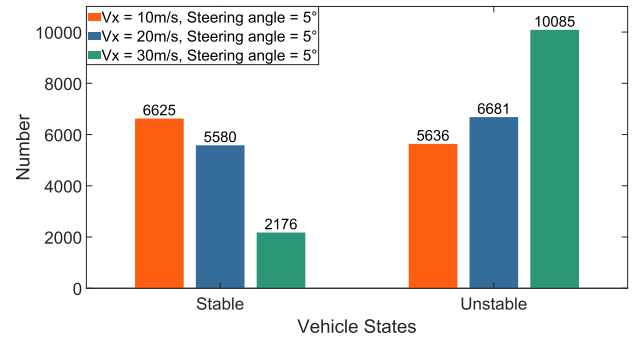


Fig. 20. Distributions of vehicle stability under different velocities with steering angle input.

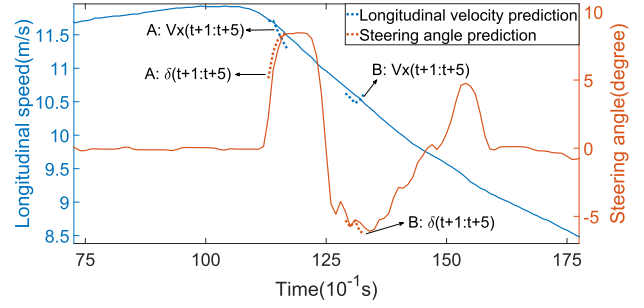


Fig. 21. State prediction sequences under evasive maneuver test.

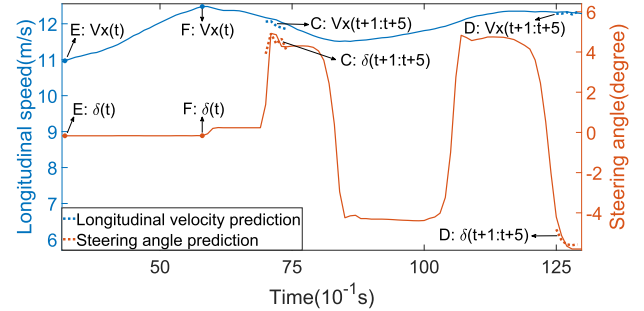


Fig. 22. State prediction sequences under pylon course slalom test.

on a low-adhesion road, for validation. These sequences are obtained using the PINN based prediction method.

Table IV and V illustrate the Lyapunov exponents of the predicted state sequences from 2 typical testing scenarios. The 3D-LEs of these sequences are all negative, indicating stable vehicle states. The magnitude of 3D-LEs represents the convergence rate of states in their respective dimensions toward their stable points. The sum of 3D-LEs characterizes the average exponential rate of change of the volume element in phase space. Experimental results illustrate that in the predicted sequences of typical test procedures, the variation of 3D-LEs accurately reflects the trend in vehicle stability.

Specifically, in sequence A of the evasive maneuver test, although the steering angle is continuously increasing, its rate of change is gradually decreasing. Meanwhile, the longitudinal speed shows a slight decline, and the yaw rate initially increases rapidly but then tends to converge to a stable point. From time step t to $t + 1$, the 3D-LEs increase by an average of 75.85% as the steering angle input increase, corresponding

TABLE IV
3-DIMENSIONAL LYAPUNOV EXPONENTS UNDER EVASIVE
MANEUVER TEST

	$[V_x, V_y, \dot{\psi}, \delta]$ (m/s, m/s, rad/s, degree)	3D-LEs ($V_y, \dot{\psi}, \dot{\phi}, \text{Sum}$)
A	$S_t: (11.74, -0.016, 0.055, 2.727)$	$(-3.439, -5.038, -12.278, -20.76)$
	$S_{t+1}: (11.71, 0.033, 0.122, 5.124)$	$(-0.296, -2.273, -2.298, -4.87)$
	$S_{t+2}: (11.71, 0.058, 0.177, 7.068)$	$(-0.173, -2.762, -2.776, -5.71)$
	$S_{t+3}: (11.55, 0.065, 0.214, 7.977)$	$(-0.185, -3.075, -3.081, -6.34)$
	$S_{t+4}: (11.40, 0.043, 0.231, 8.348)$	$(-0.192, -3.187, -3.215, -6.59)$
	$S_{t+5}: (11.28, 0.014, 0.240, 8.427)$	$(-0.196, -3.223, -3.238, -6.66)$
B	$S_t: (10.72, -0.096, -0.239, -4.751)$	$(-1.741, -3.395, -3.409, -8.55)$
	$S_{t+1}: (10.63, -0.078, -0.264, -5.289)$	$(-0.727, -2.738, -2.747, -6.21)$
	$S_{t+2}: (10.56, -0.070, -0.279, -5.767)$	$(-0.412, -2.489, -2.511, -5.41)$
	$S_{t+3}: (10.48, -0.029, -0.293, -5.298)$	$(-0.915, -2.871, -2.898, -6.68)$
	$S_{t+4}: (10.52, -0.045, -0.306, -6.096)$	$(-0.314, -2.454, -2.477, -5.25)$
	$S_{t+5}: (10.73, -0.007, -0.319, -6.281)$	$(-0.260, -2.441, -2.475, -5.18)$

to a remarkable decline in vehicle stability. Subsequently, the sum of Lyapunov exponents shows a decreasing trend, attributed to the gradual stabilization of steering angle input. It demonstrates that the variation in the Lyapunov exponents is strongly correlated with the rate of change of the steering angle input. Sequence B also demonstrates that the fluctuations of 3D-LEs accurately correspond to the changes in steering angle.

In the pylon course slalom test, sequence C exhibits a similar phenomenon. From time step $t + 2$ to $t + 5$ in sequence D, the convergence rates of vehicle states maintain a consistent level due to the minor changes in longitudinal velocity and steering angle. In addition, the sums of 3D-LEs at points E and F in table V demonstrates that as the longitudinal speed increases from 10.97 m/s to 12.43 m/s, the vehicle stability decreases by 11.88%. Therefore, in real vehicular operations, vehicle dynamics instability is strongly correlated with longitudinal motion.

F. Stability Control With Predictive Stability Assessment

In addition to validating the effectiveness of the 3D-LEs method using field test data, a stability control module is integrated into the vehicle system to demonstrate the close-loop behavior with and without the proposed method. This module employs sliding mode control to generate additional yaw moment, significantly augmenting vehicle stability under extreme operating conditions. Ensuring safety during real-vehicle testing is challenging due to the difficulty in collecting experimental data that shows vehicles becoming unstable. Therefore, unstable driving scenarios are generally simulated using a calibrated vehicle dynamics model. In this paper, an extreme driving scenario is constructed on a high-adhesion road surface. The longitudinal velocity is 30 m/s, the steering angle increases from 0° to 26° , remains constant for a period, and then decreases back to 0° , with the entire process lasting 1.5s. For ease of verification, we treat the actual sequence as the predicted sequence.

TABLE V
3-DIMENSIONAL LYAPUNOV EXPONENTS UNDER PYLON COURSE
SLALOM TEST

	$[V_x, V_y, \dot{\psi}, \delta]$ (m/s, m/s, rad/s, degree)	3D-LEs ($V_y, \dot{\psi}, \dot{\phi}, \text{Sum}$)
C	$S_t: (12.17, -0.009, 0.063, 2.162)$	$(-3.125, -6.191, -13.706, -23.02)$
	$S_{t+1}: (12.06, 0.016, 0.121, 3.908)$	$(-1.183, -3.022, -3.059, -7.26)$
	$S_{t+2}: (12.07, 0.021, 0.187, 4.922)$	$(-0.280, -2.216, -2.239, -4.74)$
	$S_{t+3}: (11.94, -0.008, 0.230, 4.410)$	$(-0.551, -2.454, -2.480, -5.49)$
	$S_{t+4}: (11.89, -0.024, 0.297, 4.657)$	$(-0.415, -2.318, -2.338, -5.07)$
	$S_{t+5}: (11.85, -0.103, 0.279, 4.035)$	$(-1.211, -3.047, -3.069, -7.33)$
D	$S_t: (12.33, -0.488, -0.030, -3.751)$	$(-1.145, -3.033, -3.033, -7.21)$
	$S_{t+1}: (12.28, -0.482, -0.027, -4.824)$	$(-0.246, -2.183, -2.225, -4.65)$
	$S_{t+2}: (12.28, -0.356, -0.190, -5.372)$	$(-0.175, -2.229, -2.264, -4.67)$
	$S_{t+3}: (12.32, -0.188, -0.292, -5.572)$	$(-0.169, -2.274, -2.291, -4.73)$
	$S_{t+4}: (12.26, -0.035, -0.329, -5.609)$	$(-0.216, -2.259, -2.273, -4.75)$
	$S_{t+5}: (12.33, 0.098, -0.341, -5.570)$	$(-0.168, -2.268, -2.282, -4.72)$
E	$S_t: (10.97, -0.041, -0.003, -0.168)$	$(-2.720, -10.417, -20.115, -33.25)$
F	$S_t: (12.43, -0.047, -0.005, -0.163)$	$(-2.818, -8.866, -17.617, -29.30)$

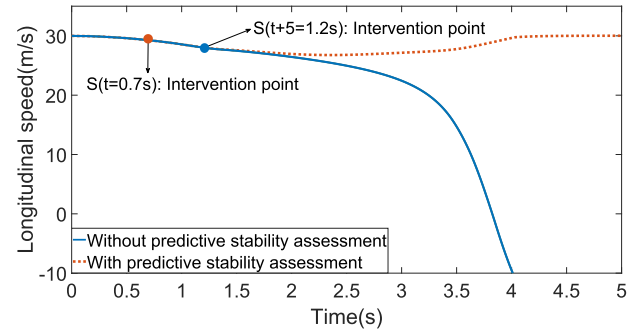


Fig. 23. Closed-loop behavior with and without predictive stability assessment method.

In table VI, the 3D-LEs of the predicted state sequence illustrate that the unstable state occurs at the time step $t + 5 = 1.2s$, where the value of 3D-LEs becomes positive. Fig. 23 shows the closed-loop behavior with and without the proposed stability assessment method. The stability control module with the predictive stability assessment intervenes 0.5s in advance, thereby avoiding the risk of vehicle instability. Without the predictive stability assessment, the stability controller intervenes at the unstable time step but fails to return the system to a stable state. This demonstrates that the stability control module with the predictive stability assessment method can achieve safer and more stable control under extreme conditions compared to the module without the proposed method.

V. CONCLUSION

To predict the potential risk of vehicle instability under extreme conditions, this paper proposes a predictive vehicle stability assessment method using 3-dimensional Lyapunov exponents (3D-LEs) for an electric vehicle. Firstly, we establish an auto-tuned nonlinear vehicle dynamics model characterized by ordinary differential equations (ODEs). Next, the

TABLE VI
3-DIMENSIONAL LYAPUNOV EXPONENTS OF PREDICTED
STATE SEQUENCES UNDER AN EXTREME DRIVING SCENARIO

$[V_x, V_y, \dot{\psi}, \delta]$ (m/s, m/s, rad/s, degree)	3D-LEs ($V_y, \dot{\psi}, \dot{\phi}, \text{Sum}$)
$S_{t=0.7s}$: (29.45,-2.45,0.57,26.05)	(-0.456,-3.264,-3.386,-7.11)
$S_{t+1=0.8s}$: (29.27,-3.05,0.56,26.14)	(-0.468,-3.068,-3.102,-6.64)
$S_{t+2=0.9s}$: (29.06,-3.62,0.55,26.21)	(-0.478,-3.048,-3.085,-6.61)
$S_{t+3=1.0s}$: (28.83,-4.18,0.55,26.08)	(-0.476,-3.025,-3.061,-6.56)
$S_{t+4=1.1s}$: (28.56,-4.74,0.55,25.22)	(-0.448,-2.867,-2.893,-6.21)
$S_{t+5=1.2s}$: (28.28,-5.30,0.56,22.54)	3D-LEs[4e3]: (1.000,0.384,-1.008,0.03) 3D-LEs[1e5]: (NaN,NaN,NaN,NaN)

[1] 3D-LEs[4e3] represents that the iteration number of 3D-LEs is 4e3, 3D-LEs[1e5] represents that the iteration number is 1e5. In this paper, the total number of iterations for the 3D-LEs calculation is 1e5.

[2] NaN represents 'not a number', indicating that the vehicle state has undergone a sudden change.

physical characteristics of vehicle system are integrated into a bidirectional long short-term memory network to construct a physics-informed neural network (PINN) for precisely predicting vehicle states, including longitudinal velocity, lateral velocity, yaw rate and steering angle. Finally, the 3D-LEs, encompassing lateral motion, yaw motion, and roll motion, are employed to comprehensively evaluate vehicle instability.

Experimental results demonstrate that the predictive vehicle stability assessment method can accurately predict vehicle motion states over multiple time steps and evaluate their potential risk of instability. Owing to the vehicle dynamic characteristics derived from ODEs, PINN based method can achieve physically consistent predictions in the longitudinal, lateral, and yaw motions. Compared to the purely data-driven method, the prediction errors of PINN based method decrease by 13.07% on average. In terms of vehicle stability assessment, the 3D-LEs based method can determine the vehicle stability boundary precisely and quantify the degree of stability. Specifically, in the absence of steering angle input, the stability region is symmetric with respect to the origin. As the speed increases, a significant contraction of the stability region is observed. With steering angle input, the stability region exhibits asymmetry due to the lateral tire forces induced by steering. In a high-speed cornering scenario, characterized by a longitudinal velocity of 30 m/s and a steering angle of 5°, the stability region decreases sharply by 61.0%.

Through a collection of typical testing scenarios on ice-snow roads, including the evasive maneuver test and pylon course slalom test, the variation in the 3D-LEs of the predicted sequences accurately reflects the trend in vehicle stability. In addition, the stability control module equipped with the predictive stability assessment method can intervene earlier, ensuring safer and more stable control under extreme operating conditions.

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